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Synthesis, crystal structure and characterization of chiral, three-dimensional anhydrous potassium tris(oxalato)ferrate(III)

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ABSTRACT

The synthesis, crystal structure and physical properties of chiral, three-dimensional anhydrous potassium tris(oxalato)ferrate(III) $[K_3F_e(C_2O_4)_3]$ are described. X-ray analysis reveals that the compound crystallized in the chiral space group $P4_132$ of cubic system with $a=b=c=13.5970(2)$, $Z=4$. The structure of the complex consists of infinite anionic $[Fe(C_2O_4)_3]^{3-}$ units with distorted octahedral environment of iron surrounded by six oxygen atoms of three oxalato groups. The anionic units are interlinked through K^+ ions of three different coordination environments of distorted octahedral, bicapped trigonal prismatic and trigonal prismatic yielding a three-dimensional motif. The two broad absorption bands at 644 and 924 nm from UV–vis–NIR transmittance spectra were ascribed to a ligand-to-metal charge transfer. The room temperature crystalline EPR spectra indicate the high-spin ($S=5/2$) of Fe(III) ion. The vibrating sample magnetometer measurement shows the paramagnetic nature at room temperature. Thermal studies of the compound confirm the absence of water molecule.

1. Introduction

The current challenge in the field of material science is synthesis and study of new materials with multifunctional properties due to their potential applications in molecular magnets, chiral materials, superconductors, ferromagnetic metals and photophysics [1,2]. In this regard, the charge-transfer salts of tris(oxalato)metalate(III) anions are suitable for building multifunctional properties in a molecular lattice [3–5]. The flexible bonding mode of oxalato ion with metal ions and its ability to transmit electronic effects when acting as bridges between paramagnetic centres and also synthetic control over the lattice dimensionality enables to design two- and three-dimensional multifunctional magnets [6,7]. Introducing chirality into these materials is expected to induce chiral spin structures and also display magneto-chiral dichroism [8–11]. The tris(oxalato)ferrate anion is an interesting "building block" due to its magnetic properties and possibility of different dimensional network formations, moreover tris(oxalato)ferrate(III) salts are important precursors for the synthesis of nanoferrites [12,13]. The objective of the present work is to prepare a chiral, three-dimensional tris(oxalato)ferrate(III) multifunctional material by using subtle changes in the synthesis procedure.

The synthesis of potassium tris(oxalato)ferrate(III) trihydrate was reported for the first time by Blair and Jones in 1939 [14] and it belongs to monoclinic system with $P2_1/c$ space group [15,16]. This material was extensively investigated owing to its wide applications in photochemical studies, actinometery, sensors, magnetic materials, etc. [17–19]. We have synthesized anhydrous potassium tri(oxalato)ferrate(III) compound by a modified method. Herein, we report the results on synthesis, crystal structure, optical, thermal and magnetic properties of chiral, three-dimensional anhydrous potassium tris(oxalato)ferrate(III) complex.

2. Experimental

2.1. Synthesis and crystal growth

All the chemicals were of reagent grade and used without further purification. The anhydrous potassium tris(oxalato)ferrate(III) compound was prepared by a modified procedure different from that reported in literature [14]. The crystals of $K_3[Fe(C_2O_4)_3]$ were obtained by slow evaporation of a mixture of solutions obtained from freshly prepared $Fe(OH)_3$ solid and the aqueous solution of $K_2C_2O_4 \cdot H_2O$ and $H_2C_2O_4 \cdot 2H_2O$.

The $Fe(OH)_3$ was obtained by dissolving (1.112 g, 2 mmol) $FeSO_4 \cdot 7H_2O$ in 15 ml of water then by adding 1 ml concentrated HNO_3 and boiled for about 5 min for the removal of excess HNO_3 as NO_2 fumes. The solution was diluted with water and 5 ml 20%

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Fig. 1. Photograph of $K_3[Fe(C_2O_4)_3]$crystals.

NaOH aqueous solution was added. On centrifugation a reddish brown precipitate of $Fe(OH)_3$ solid was formed. A 50 ml of aqueous solution was prepared on dissolving (1.104 g, 3 mmol) $K_2C_2O_4 \cdot H_2O$ and (0.756 g, 3 mmol) $H_2C_2O_4 \cdot 2H_2O$. $Fe(OH)_3$ solid was gradually added to this aqueous solution on constant stirring. The solution was boiled for about 15 min and filtered at cold condition to remove unreacted $Fe(OH)_3$. The green solution was kept at room temperature for three days to get crystals. The grown crystals are shown in Fig. 1.

2.2. Characterization techniques

Intensity data for the anhydrous potassium tris(oxalato)ferrate(III) $[K_3[Fe(C_2O_4)_3]$ crystal was collected on Oxford X Calibur, Gemini diffractometer equipped with EOS CCD detector at 293 K. Monochromatic Mo-$K\alpha$ radiation (λ=0.71073 Å) was used for the measurements. Data were collected and reduced by using the "CrysAlispro" program [20]. An empirical absorption correction using spherical harmonics was implemented in "SCALE3 ABSPACK" scaling algorithm. The crystal structure was solved by direct methods using SHELXS-97 [21] and the refinement was carried out against F^2 using SHELXL-97. All non-hydrogen atoms were refined anisotropically. Elemental analysis was performed with a Elementar Vario Micro cube CHN analyzer. UV–vis–NIR spectra are obtained from JASCO V-570 spectrophotometer. ESR measurements were performed with a JEOL JES-FA200 spectrometer on a single crystal at room temperature. The magnetization measurement was performed on a crystalline sample by Lakeshore Vibrating Sample Magnetometer at room temperature. The thermal stability of the compound is tested by using TA Q20 differential scanning calorimeter under a nitrogen atmosphere with a heating rate of 10 °C/min.

3. Results and discussion

The elemental analysis of the crystal reveals that it contains carbon 16.42% (16.47%), and it does not contain hydrogen.

The compound anhydrous potassium tris(oxalato)ferrate(III) crystallizes in cubic chiral space group $P4_132$ with three-dimensional arrangement which is a rare case in inorganic chemistry [3,5]. Table 1 shows the crystallographic data at 293 K. A similar crystal structure is observed in case of potassium sodium tris(oxalato)ferrate(III) $K_{2.5}Na_{0.5}[Fe(C_2O_4)_3]$ in which 1/6 of K^+ ions are replaced by Na^+ ions [22,23]. Fig. 2 shows the standard ORTEP diagram with the atom numbering scheme and 50% thermal ellipsoid. The atomic coordinates and equivalent isotropic displacement parameters are listed in Table 2. The CCDC 825674 contains the supplementary crystallographic data.

In the crystal structure the central iron atom is in a distorted octahedral environment, being co-ordinated to six oxygen atoms of three oxalato groups. The distortion arising because of the reduced bite angles of the oxalate ligand 180° for O2(vi)–Fe1–O4(vi), 87° for O2(viii)–Fe1–O2(vii), 88° for O4(vi)–Fe1–O4(viii), 108° for O2(vii)–Fe1–O4(vi) and O2(viii)–Fe1–O4(vii), and 159° for O2(viii)–Fe1–O4(vi) (Fig. 3). The three oxalato groups adopt chelating coordination mode and act as bidentate through O(2) and O(4) towards the iron atom. The Fe–O_{ox} bond lengths range from 2.011(4) Å to 2.033(4) Å (Table 3). These bond lengths are in good agreement with those of other potassium tris(oxalato) metal complexes [24,25].

The potassium cations K(1), K(2) and K(3) adopt three different types of coordination with oxygen atoms of oxalato ions (Fig. 4).

The ion K(1) coordinate with six oxygen atoms of the oxalato ions [three O(1) and three O(2)] as a distorted octahedra. The ion K(2) coordinate with eight oxygen atoms [two O(1), two O(3) and four O(4)] as a bicapped trigonal prismatic and ion K(3) coordinate with six O(3) oxygen atoms as a trigonal prismatic. In these three geometries the K–O distances are different from each other. In the distorted octahedral geometry of K(1) the bond lengths of the three K(1)–O(1) and three K(1)–O(2) are 2.703(4) Å and 3.000(5) Å respectively. In the bicapped trigonal prismatic geometry of

Table 1
Crystal data and structure refinement for anhydrous $K_3[Fe(C_2O_4)_3]$.

Empirical formula	$C_{12}Fe_2K_6O_{24}$
Formula weight	874.42
Temperature	293(2) K
Wavelength	0.71073 Å
Crystal system	Cubic
Space group	$P4_132$
Unit cell dimensions	a=13.5970(2) Å $\alpha=\beta=\gamma=90°$.
Volume	2513.79(6) $Å^3$
Z	4
Density (calculated)	2.310 Mg/m^3
Absorption coefficient	2.259 mm^{-1}
F(000)	1720
Crystal size	$0.40 \times 0.38 \times 0.32$ mm^3
Theta range for data collection	3.35°–26.33°
Index ranges	$-9<=h<=7$, $-8<=k<=16$, $-8<=l<=16$
Reflections collected	2211
Independent reflections	835 [R(int)=0.0249]
Completeness to theta=25.00°	97.1%
Absorption correction	Semi-empirical from equivalents
Max. and min. transmission	0.5317 and 0.4652
Refinement method	Full-matrix least-squares on F^2
Data/restraints/parameters	835/0/69
Goodness-of-fit on F^2	1.162
Final R indices [I > 2sigma(I)]	$R1$=0.0395, w$R2$=0.1120
R indices (all data)	$R1$=0.0430, w$R2$=0.1145
Absolute structure parameter	−0.07(6)
Extinction coefficient	0.0087(16)
Largest diff. peak and hole	0.977 and −1.584 e $Å^{-3}$

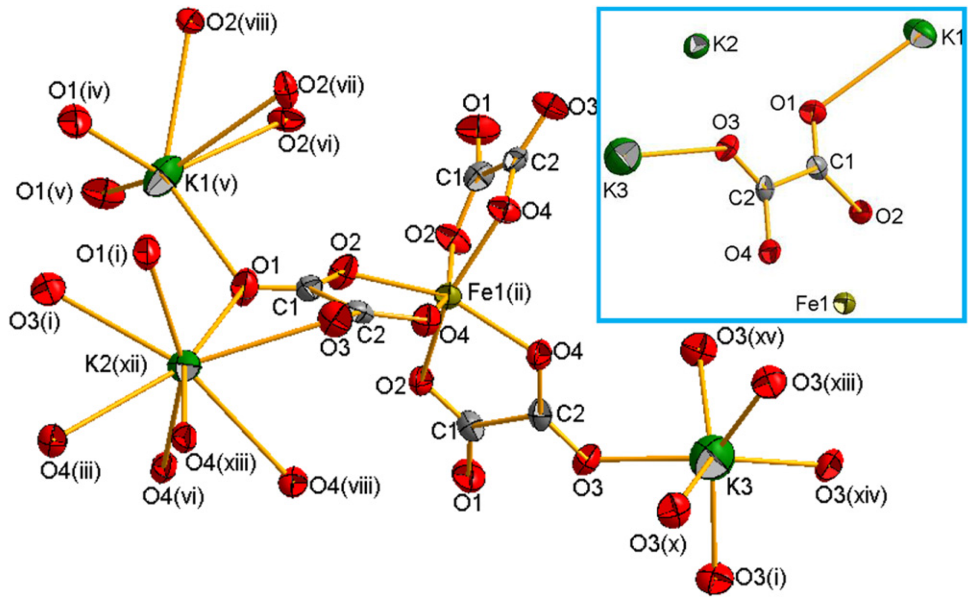

Fig. 2. Thermal ellipsoidal plot of $K_3[Fe(C_2O_4)_3]$ **(1) basic** unit **(shown in inset)** with atom labeling scheme. Displacement ellipsoids are drawn at 50% probability level. The complete coordination sphere of **Fe(1), K(1), K(2) and K(3)** generated through symmetry relation and the relevant symmetry codes are: (i) $1.5-y$, $1-z$, $-0.5+x$; (ii) $0.25+y$, $1.25-x$, $-0.25+z$; (iii) $1-y$, $0.5+z$, $0.5-x$; (iv) $0.5+y$, $0.5-z$, $1-x$; (v) $1-z$, $-0.5+x$, $0.5-y$; (vi) $0.75-z$, $0.75-y$, $0.75-x$; (vii) $0.25+x$, $0.25-z$, $-0.25+y$; (viii) $1.25-y$, $-0.25+x$, $0.25+z$; (ix) $1.5-z$, $1-x$, $-0.5+y$; (x) $0.5+z$, $1.5-x$, $1-y$; (xi) $0.25+z$, $1.25-y$, $-0.25+x$; (xii) $0.5-z$, $1-x$, $-0.5+y$; (xiii) $-0.25+y$, $0.25+x$, $0.25-z$; (xiv) $0.75-z$, $1.75-y$, $0.75-x$ and (xv) $1.25-x$, $0.75+z$, $-0.75+y$.

Table 2
Atomic coordinates ($x \times 10^4$) and equivalent isotropic displacement parameters U_{eq} ($Å^2 \times 10^3$) for $K_3[Fe(C_2O_4)_3]$. U_{eq} is defined as one third of the trace of the orthogonalized U^{ij} tensor.

Atom	x	y	z	U_{eq}
C(1)	7066(3)	5506(3)	194(3)	24(1)
C(2)	6412(3)	6377(3)	577(3)	22(1)
O(1)	7960(3)	5607(3)	143(3)	38(1)
O(2)	6579(2)	4741(2)	−42(3)	30(1)
O(3)	6799(3)	7090(3)	958(3)	33(1)
O(4)	5475(2)	6222(2)	469(3)	27(1)
K(1)	9204(1)	4204(1)	796(1)	46(1)
K(2)	5071(1)	6250	2571(1)	31(1)
K(3)	6250	8750	1250	61(1)
Fe(1)	7609(1)	2609(1)	2391(1)	21(1)

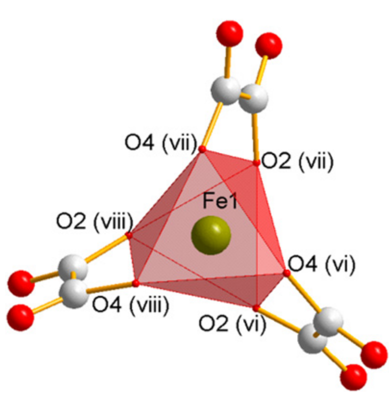

Fig. 3. Octahedral coordination geometry around iron in $K_3[Fe(C_2O_4)_3]$ with relevant symmetry codes (vi) $0.75-z$, $0.75-y$, $0.75-x$; (vii) $0.25+x$, $0.25-z$, $-0.25+y$; and (viii) $1.25-y$, $-0.25+x$, $0.25+z$.

K(2) the bond lengths of the two K(2)–O(1) and the two K(2)–O(3) are 2.698(5) Å and 2.949(5) Å respectively. The bond lengths of other four K(2)–O(4) are within the range of 2.920(5) to 3.065(4) Å respectively. The bond lengths of other four K(2)–O(4) are within the range of 2.920(5)–3.065(4) Å. In the trigonal prismatic geometry of K(3) the bond lengths of all the six K(3)–O(3) are 2.409(4) Å.

The UV–vis–NIR spectrum of $K_3[Fe(C_2O_4)_3]$ exhibits two broad absorption bands at 644 and 924 nm (Fig. 5) attributed to $^6A_1 \rightarrow {}^4T_2$ and $^6A_1 \rightarrow {}^4T_1$ ligand-to-metal charge transfer in distorted octahedral Fe(III) complex with high spin state [26–29].

The X-band (9.45 GHz) ESR spectra of $K_3[Fe(C_2O_4)_3]$ at room-temperature (Fig. 6) show an intense resonance signal with both peak to peak line width of 55.96 mT around $g=5.8$. The features of the ESR spectrum are very similar to those reported by Collison and Powell for its trihydrate counterpart [30].

The variation of induced magnetic moment with respect to the applied magnetic field would be identified by vibrating sample magnetometer for the analysis of magnetic nature of the material. The observed magnetization nature of the compound is shown in Fig. 7. It is observed that the intensity of magnetization varies linearly with the applied field, which shows the paramagnetic nature at room temperature.

Table 3
Selected bond lengths (Å) and angles (deg.) for anhydrous $K_3[Fe(C_2O_4)_3]$.

C1–O1	1.222(7)	O1(iv)–K1–O2(vi)	135.21(15)
C1–O2	1.275(7)	O1–K1–O2(vi)	90.55(13)
C1–C2	1.571(8)	O1(iv)–K1–O2(vii)	82.71(14)
C2–O3	1.212(7)	O1(iv)–K1–Fe1	103.50(12)
C2–O4	1.301(7)	O1(v)–K1–K2(ix)	85.36(12)
O1–K2(i)	2.698(5)	O1–K1–K2(ix)	119.84(13)
O1–K1	2.703(4)	O1(x)–K2–O1(viii)	95.0(2)
O2–Fe1 ii)	2.011(4)	O1(x)–K2–O4	153.78(12)
O2–K1(ii)	3.000(5)	O1(viii)–K2–O4	82.05(14)
O3–K3	2.409(4)	O1(viii)–K2–O3(x)	88.98(13)
O3–K2(i)	2.949(5)	O1(x)–K2–O4(xii)	149.69(12)
O3–K2	3.415(5)	O1(viii)–K2–O4(xii)	86.27(13)
O4–Fe1(ii)	2.033(4)	O1(x)–K2–O3	115.45(14)
O4–K2	2.920(5)	O2(vi)–K1–K2(ix)	149.57(9)
O4–K2(iii)	3.065(4)	O2(vii)–K1–K2(ix)	97.41(8)
K1–O1(iv)	2.703(4)	O2(viii)–K1–K2(ix)	122.91(8)
K1–O1(v)	2.703(4)	O2(viii)–Fe1–O2(vii)	87.01(17)
K1–O2(vi)	3.000(5)	O2(viii)–Fe1–O4(vi)	159.22(16)
K1–O2(vii)	3.000(5)	O2(vii)–Fe1–O4(vi)	108.42(17)
K1–O2(viii)	3.000(5)	O2(vi)–Fe1–O4(vi)	80.20(16)
K1–Fe1	3.758(3)	O2(viii)–Fe1–O4(vii)	108.42(17)
K2–O1(x)	2.698(5)	O3(x)–K2–O3(viii)	133.25(19)
K2–O1(viii)	2.698(5)	O3(x)–K2–O4(xii)	151.46(11)
K2–O3(x)	2.949(5)	O3(viii)–K2–O3	117.18(14)
K2–O3(viii)	2.949(5)	O3–K2–O3(xi)	176.63(17)
K2–O4(xi)	2.920(5)	O3(xiv)–K3–O3(xv)	85.47(16)
K2–O4(xii)	3.065(4)	O3(i)–K3–O3(xv)	127.3(2)
K2–O4(xiii)	3.065(4)	O3(xiv)–K3–O3	140.6(2)
K2–O3(xi)	3.415(5)	O4(xii)–K2–O4(xiii)	107.59(17)
K3–O3(xiv)	2.409(4)	O4(xi)–K2–O3	142.32(13)
K3–O3(i)	2.409(4)	O4(xii)–K2–O3	91.89(11)
K3–O3(xv)	2.409(4)	O4(xiii)–K2–O3	90.10(11)
K3–O3(x)	2.409(4)	O4(vi)–Fe1–O4(viii)	88.35(18)
K3–O3(xiii)	2.409(4)	O4(vi)–Fe1–K1	126.43(12)
Fe1–O2(viii)	2.011(4)	O4–K2–O4 xi)	111.68(17)
Fe1–O2(vii)	2.011(4)	O4–K2–O3(x)	95.04(11)
Fe1–O2(vi)	2.011(4)	O4(xi)–K2–O3(x)	110.96(11)
Fe1–O4(vi)	2.033(4)	O4(xi)–K2–O4(xii)	83.66(12)
Fe1–O4(viii)	2.033(4)	Fe1(ii)–O2–K1(ii)	95.15(15)
Fe1–O4(vii)	2.033(4)	Fe1(ii)–O4–K2	109.94(17)
K3–O3–K2(i)	101.57(15)	Fe1(ii)–O4–K2(iii)	104.78(15)
K3–O3–K2	89.61(13)	Fe1–K1–K2(i)	127.20(3)
O1(iv)–K1–O1(v)	114.73(9)		

Symmetry transformations used to generate equivalent atoms: (i) $1.5-y, 1-z, -0.5+x$; (ii) $0.25+y, 1.25-x, -0.25+z$; (iii) $1-y, 0.5+z, 0.5-x$; (iv) $0.5+y, 0.5-z, 1-x$; (v) $1-z, -0.5+x, 0.5-y$; (vi) $0.75-z, 0.75-y, 0.75-x$; (vii) $0.25+x, 0.25-z, -0.25+y$; (viii) $1.25-y, -0.25+x, 0.25+z$; (ix) $1.5-z, 1-x, -0.5+y$; (x) $0.5+z, 1.5-x, 1-y$; (xi) $0.25+z, 1.25-y, -0.25+x$; (xii) $0.5-z, 1-x, -0.5+y$; (xiii) $-0.25+y, 0.25+x, 0.25-z$; (xiv) $0.75-z, 1.75-y, 0.75-x$ and (xv) $1.25-x, 0.75+z, -0.75+y$.

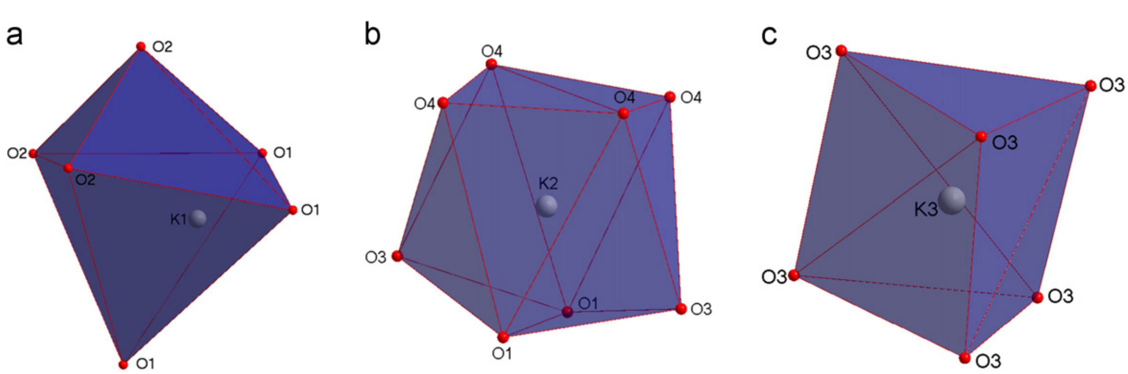

Fig. 4. (a) Distorted octahedral, (b) bicapped trigonal prism and (c) trigonal prismatic environments of K(1), K(2) and K(3) respectively in anhydrous $K_3[Fe(C_2O_4)_3]$.

The thermal studies of the compound are performed using differential scanning calorimetry in the temperature range 50–450 °C (Fig. 8). It can be seen that there is no dehydration endothermic peak around 100 °C and confirms the absence of water molecule in the compound. First endothermic peak is observed at 361 °C and second and third endothermic peaks are

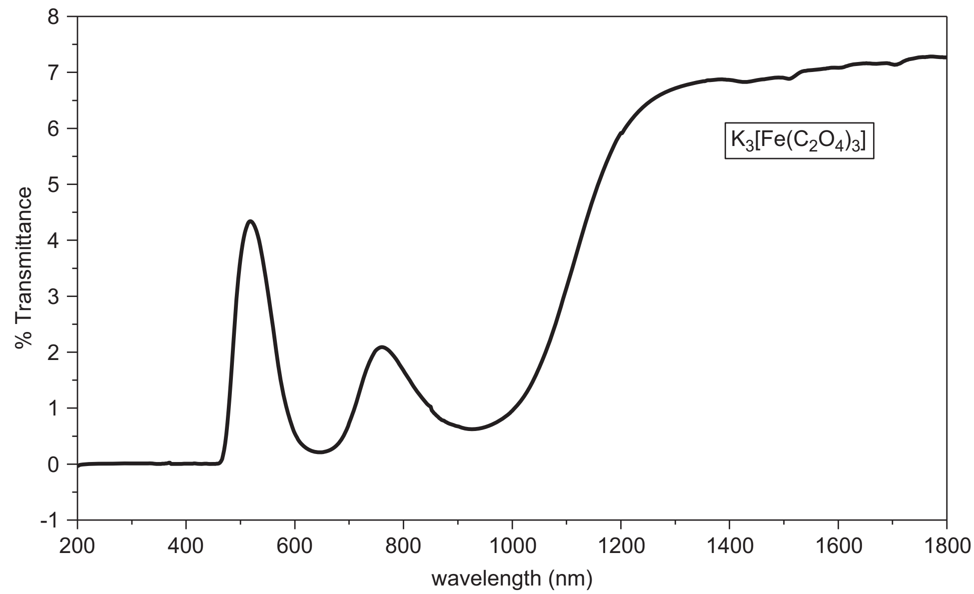

Fig. 5. UV–vis–NIR spectrum of $K_3[Fe(C_2O_4)_3]$.

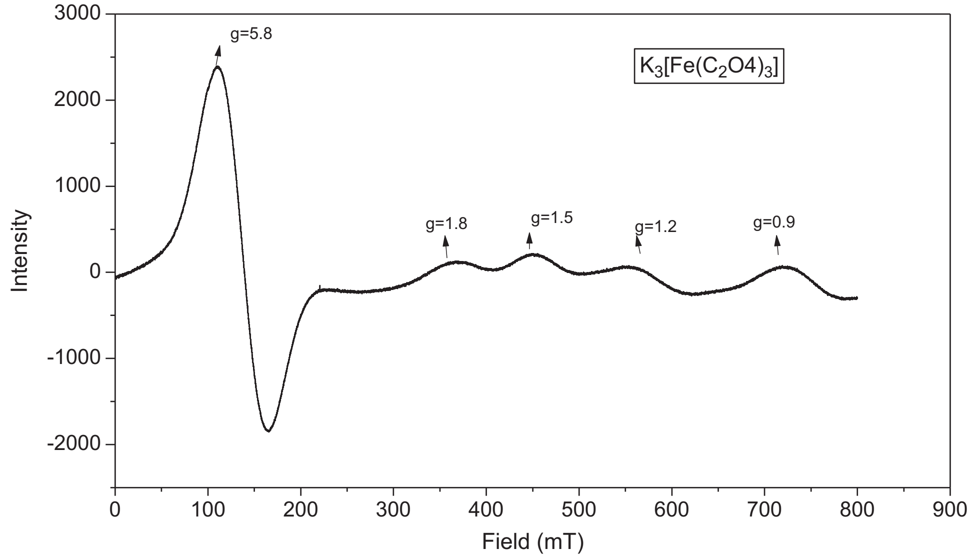

Fig. 6. X-band ESR spectra of $K_3[Fe(C_2O_4)_3]$ at room temperature.

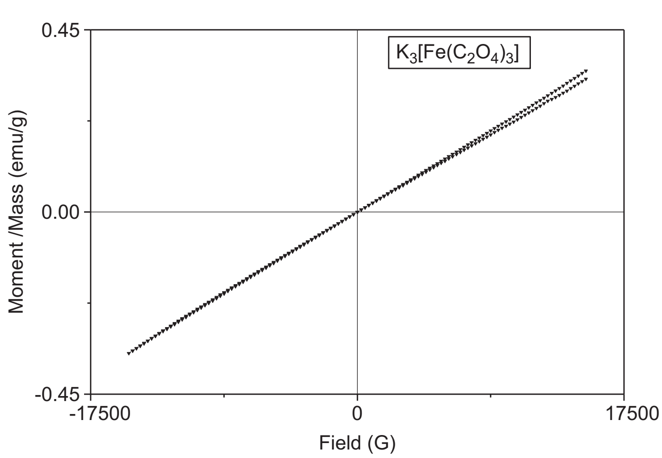

Fig. 7. The *M–H* plot of $K_3[Fe(C_2O_4)_3]$ at room temperature.

observed at 397 and 417 °C respectively. The first endothermic peak corresponds to the decomposition of the iron to its oxide and the high temperature endothermic peaks correspond to the decomposition of potassium oxalate [31,32].

4. Conclusions

The synthesized single crystals of anhydrous potassium tris (oxalato)ferrate(III) were grown from solutions on slow evaporation. The crystallographic structure of this complex is cubic with chiral space group $P4_132$ and there are four molecules per unit cell. It has been found that the structure of anhydrous complex is distinctly different with its trihydrate counterpart. Chirality and three-dimensional structure of this compound opens the way for numerous interesting optical, electrical and magnetic properties. Further low temperature magnetic studies are in progress.

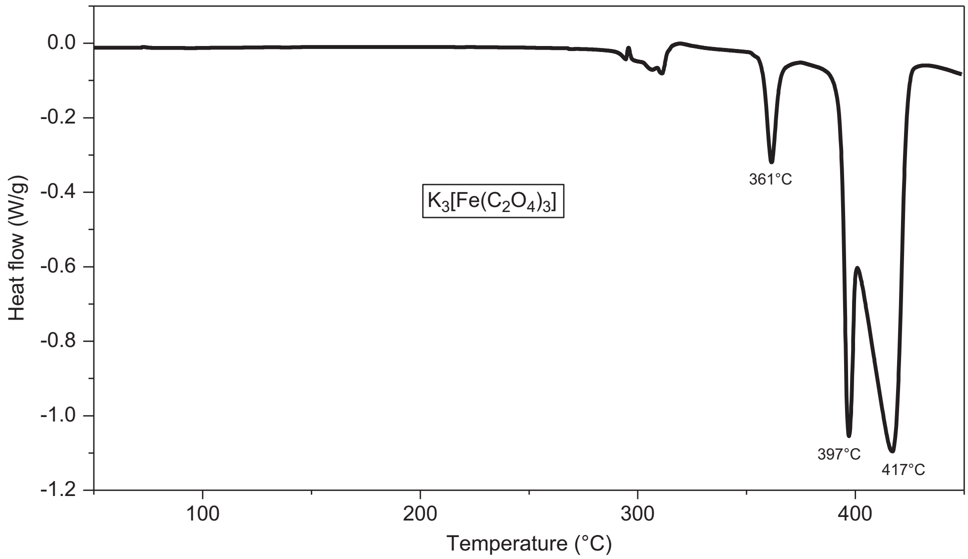

Fig. 8. DSC curve of $K_3[Fe(C_2O_4)_3]$.

Supplementary data

CCDC 825674 contains the supplementary crystallographic data. This data can be obtained free of charge via http://www.ccdc.cam.ac.uk/conts/retrieving.html, or from Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 IEZ, UK; fax: +44 1223-336-033; or e-mail: deposit@ccdc.cam.ac.uk.

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References

[1] D. Maspoch, D. Ruiz-Molina, J. Veciana, Chem. Soc. Rev. 36 (2007) 770.
[2] K.S. Gavrilenko, Y.L. Gal, O. Cador, S. Golhen, L. Ouahab, Chem. Commun. (2007) 280.
[3] S. Decurtins, H.W. Schmalle, P. Schneuwly, H.R. Oswald, Inorg. Chem. 32 (1993) 1888.
[4] S. Decurtins, H.W. Schmalle, P. Schneuwly, J. Ensling, P. Guetlich, J. Am. Chem. Soc. 116 (1994) 9521.
[5] S. Decurtins, H.W. Schmalle, R. Pellaux, P. Schneuwly, A. Hauser, Inorg. Chem. 35 (1996) 1451.
[6] S. Decurtins, H. Schmalle, H.R. Oswald, A. Linden, J. Ensling, P. Gutlich, A. Hauser, Inorg. Chim. Acta. 216 (1994) 65.
[7] Y.T. Li, C.W. Yan, H.S. Guan, Polyhedron 22 (2003) 3223.
[8] C. Train, R. Gheorghe, V. Krstic, L.M. Chamoreau, N.S. Ovanesyan, G.L.J.A. Rikken, M. Gruselle, M. Verdaguer, Nature 7 (2008) 729.
[9] G.L.J.A. Rikken, E. Raupach, Nature 390 (1997) 493.
[10] G.L.J.A. Rikken, E. Raupach, Nature 405 (2000) 932.
[11] M. Gruselle, R. Andres, B.M.M. Brissard, C. Train, M. Verdaguer, Chirality 13 (2001) 712.
[12] T.M. Muziol, G. Wrzeszcz, L. Chrzaszcz, Polyhedron 30 (2011) 169.
[13] R. Prasad, Sulaxna, A. Kumar, J. Therm. Anal. Calorim. 81 (2005) 441.
[14] J.C. Blair, E.M. Jones, in: H.S. Booth (Ed.), Inorganic Synthesis, vol. 1, McGraw-Hill Book Co., New York, 1939, p. 35, in:.
[15] P.C. Junk, J. Coord. Chem. 58 (2005) 355.
[16] G. Delgado, A.J. Mora, V. Sagredo, Physica B 320 (2002) 410.
[17] H.E. Spencer, M.W. Schmidt, J. Phys. Chem. 75 (1971) 2986.
[18] D. Armenatano, G. De Munno, F. Lioret, M. Julve, Cryst. Eng. Commun. 7 (2005) 57.
[19] Tiexiang Fu, Jun Tao, Sens. Actuat. B 129 (2008) 339.
[20] Oxford Diffraction, CrysAlis PRO, Oxford Diffraction Ltd., Yarnton, England, 2009.
[21] G.M. Sheldrick, Acta Crystallogr. A 64 (2008) 112.
[22] S. Henneicke, R. Wartchow, Z. Kristallogr. 212 (1997) 56.
[23] R. Wartchow, Z. Kristallogr 212 (1997) 57.
[24] R. Kaziro, T.W. Hambly, R.A. Binstead, J.K. Beattie, Inorg. Chim. Acta 164 (1989) 85.
[25] B.C. Dalzell, K. Eriks, J. Am. Chem. Soc. 93 (45) (1971) 4298.
[26] S.K. Gupta, Y.S. Kushwah, Polyhedron 20 (2001) 2019.
[27] D.M. Sherman, T.D. Waite, Am. Mineral. 70 (1985) 1262.
[28] T.S. Piper, R.L. Carlin, J. Chem. Phys. 35 (1961) 1809.
[29] H.E. Spencer, J. Phys. Chem. 73 (1969) 2316.
[30] D. Collision, A.K. Powell, Inorg. Chem. 29 (1990) 4735.
[31] D. Broadbent, D. Dollimore, J. Dollimore, J. Chem. Soc. A (1966) 278.
[32] P.J. Haines, Thermochim. Acta 285 (1999) 340.